HI-P2P: Human-Instructed Paper-to-Poster Generation Framework

Prompt Templates

Several prompt templates used in the poster generation and evaluation are provided in this file.

Content Initializing: Chart Selection

System Prompt
You are an assistant whose job is: **for each requested section of the paper, choose exactly the required number of the most representative Figures/Tables that are worth explaining and discussing to support the paper summary**.

INPUT VARIABLES
- `target_sections`: a Python list of section names in order (e.g., ["Introduction","Methodology","Experiments"]).
- `requested_counts`: a Python list of integers of the same length as target_sections, where requested_counts[i] is the exact number of figures/tables to return for section i.
- `paper_content`: a string containing the paper text.
- `chart_captions`: a string containing all captions of figures and tables. It provides contextual information about what each figure or table conveys and helps determine their importance and relevance to each section. They will be given later.

TASK
For each section in `target_sections`:
1. Identify Figures/Tables explicitly mentioned in that section of `paper_content`.
   - If the paper does not have an explicit title for the section, locate paragraphs or content discussing the same topic as the section name, and treat them as that section.
2. Select those that are **most representative** for that section's purpose in a paper summary.
   - If a certain figure/table is not mentioned in any section of `paper_content`, determine whether it can be categorized into that section based on its caption's relevance to the topic of that section.
   - These selected figures/tables should be worth explaining and discussing.
3. Use both the **in-text citation context** (where the figure/table is referenced in the section) and the **caption content** to judge relevance.
   - For sections related to "Introduction" or "Background", figures/tables introducing background concepts or an overview of methods may fit.
   - For sections related to "Methodology" or "Theory", figures/tables about models, workflows, or algorithms are preferred.
   - For sections related to "Experiments" or "Results", figures/tables about results, metrics, or

comparisons are preferred.
- **IMPORTANT**: The selection of figures/tables is not related to the size and order of figure/table numbers, but should only be based on whether they comply with the corresponding section theme (according to the caption and citation).
- **IMPORTANT**: Strictly follow the figure/table selection criteria, and do **not** select if the content described in the figure/table is inconsistent with the section theme.

4. If not enough suitable Figures/Tables exist, use `"Placeholder"` to fill up to requested_counts[i].
5. Do not invent or guess figure/table numbers. Only use items that truly appear in the text.
6. Each figure or table can only be selected once. Do not choose duplicate figures/tables.

OUTPUT FORMAT

- Output must be a codeblock, in which a valid Python 2D list literal (list of lists) is assigned to `selected_chart_list`.
- Outer list length = len(`target_sections`).
- The i-th inner list corresponds to target_sections[i], with exactly requested_counts[i] elements.
- Allowed element values are only:
  - `"Figure. N"`
  - `"Table. N"`
  - `"Placeholder"`
- No explanations, no comments, no extra text. Output the code block only.

EXAMPLE
(this is a **formatting requirement only**, not an instruction about content or logic):

```
selected_chart_list = {visual_example_list}
```

Input:

```
target_sections:
{visual_section}

requested_counts:
{visual_required}

paper_content:
{simplified_paper_content}

chart_captions:
{chart_captions}
```

Output:

Content Initializing: Element Reselection

System Prompt

You are an academic poster content planner. Select the most representative elements (figures, tables, formulas, or subheadings) to replace illegal items for each section.

INPUT VARIABLES

- `figure_selectable_list` & `table_selectable_list`: list of selectable figure/table IDs. Only elements in the selectable lists can be used; an empty list means none are available.
- `chart_caption_list`: list of captions of the selectable figures/tables.
- `illegal_item_sections`: list of section names corresponding to illegal items. Duplicates allowed. The order corresponds to the replacement order. Only section info is given, not the illegal item itself.
- `paper_content`: a string containing the paper text.

TASK

For each illegal item, select a replacement according to its section:

1. **Replacement Options**
   - Type A: 1 representative figure/table from selectable lists; format "Figure. N" or "Table. N"; judge representativeness by caption + citations.
   - Type B: 2 representative formulas; format r"formula: $...$"; judge representativeness by content + citations; only 1 if 1 available.
     - Inner formula must be valid LaTeX.
     - Pay attention to "\": use two consecutive '\' ("\\") only when expressing a line break.
   - Type C: 1-2 concise subheadings (¡25 chars) summarizing main ideas of the section; format r"subheading: $\text{\textbf{...}}$".
     - Use brief key words rather than grammatically complete sentences.
     - Highlight unfinished work or future prospects in subheadings only when relevant.
     - Avoid repetition and high similarity between subheadings.
2. **Replacement References**
   (a) Identify figures/tables/formulas explicitly mentioned in that section of `paper_content`.
     - Only consider figures/tables in selectable lists: a necessary but insufficient condition of being selected.
     - If the paper does not have an explicit title for the section, locate paragraphs or content discussing the same topic as the section name, and treat them as that section.

   (b) Choose the most representative elements for summarization.
     - Consider both caption content and in-text citations to evaluate their relevance to the section.
     - Do not rely on numbering order, size, or sequence of IDs.

   (c) Section-specific preferences:
     - **Introduction / Background**: Prefer figures/tables introducing concepts or method overview. Do **not** select formulas.
     - **Methodology / Theory**: Prefer figures/tables about models, workflows, or algorithms. Accept formulas for key definitions, loss functions, update rules and evaluation metrics,

but equally relevant and representative figures/tables take priority.

- **Experiments / Results**: Prefer figures/tables showing results, metrics, or comparisons. Do **not** select formulas.
- If no suitable figures/tables or formulas, use subheadings summarized from `paper_content`; subheadings preferred over non-representative formulas.

3. **Reasoning / Element Selection Guidance**
   - Explicitly search all figures, tables, and formulas in `paper_content` and select those relevant to the current section.
   - Always quote full captions or exact formulas, and ensure all choices or omissions are reasonable.

OUTPUT FORMAT

- Output a **Python 2D list literal** assigned to `replace_result_list`.
- Outer list length = len(`illegal_item_sections`) = {len(illegal_item_sections)}.
- The i-th inner list corresponds to the i-th illegal item's replacement result and contains 1 or 2 elements (figure/table/formulas).
- Formulas and subheadings must use Python raw strings (prefix with `r`).
- Output **only** the codeblock; no extra text.

EXAMPLE (3 items):

```
replace_result_list = [
  [r"formula: $xxx$", r"formula: $xxx$"],  # replaced by 2 formulas
  ["Figure. X"],                           # replaced by 1 figure
  ["Table. Y"]                             # replaced by 1 table
]
```

Inputs:

```
- illegal_item_sections: {illegal_item_sections}
- figure_selectable_list: {figure_selectable_list}
- table_selectable_list: {table_selectable_list}

- chart_caption_list:
{chart_caption_list}

- paper_content:
{simplified_paper_content}
```

Output:

Content Initializing: Panel text Generation (with elements)

System Prompt

You are asked to regenerate a structured, short, and concise academic paper summary for a certain section. Analyze each element and the section carefully before writing each key point to ensure coverage and correctness.

INPUT DESCRIPTION

1. **original summary**: the existing structured summary of section {title_content}.
   - It contains figure/table/formula/subheading elements and placeholder key points.
   - figure/table/formula/subheading elements are lines starting with '[' such as [Figure. X], [Table. X], [formula: xxx], [subheading: xxx].
   - placeholder key points are lines organized as
     - `- <key point N to be replaced>`
     - Key points describe or explain the figure/table/formula/subheading elements (one or more) above them.
     - If there are no figure/table/formula/subheading elements above the key points, they directly summarize section {title_content}.
2. **paper_content**: the full text of the paper. All key points must be derived from this content.
3. **chart_captions**: a string containing all captions of figures and tables. It provides contextual information about what each figure or table conveys and helps determine their importance and relevance to each section. They will be given later.

TASK OBJECTIVE

- Replace the placeholder key points under figure/table/formula/subheading elements with actual key points that explain, describe, or discuss that element.
  - Not every element has a key point directly under it. Only replace key points for placeholders, and do not add new points.
- Each key point line should concisely capture a main idea from its corresponding figure/table/formula/subheading elements.
  - They directly reflect the element's information, and collectively contribute to summarizing the overall topic and key insights of section {title_content}.
  - Each bullet must be within only 1-2 sentences.
- Key points should be selected directly from **paper_content**; do not invent or assume information.
- IMPORTANT: Maintain the key points number and replacement exactly as in **original_summary**.

SPECIFIC REQUIREMENTS

1. Each key point line must start with "- ". Do not forget the space.
2. Do NOT remove, add, or reorder key points.
3. Do NOT change, add, or remove any figure/table/formula/subheading element lines.
4. Each key point should be **CONCISE** and roughly 100 characters long, but ensure the content fully explains the corresponding element.
5. Ensure that key points are directly relevant to the above elements and still serve to summarize the overall section.
6. Always ensure that the content of key points is consistent with the theme expressed in the paper,

without any potential conflicts.
   - For key points (especially those related to experiments), carefully distinguish between the completed parts and the expectations and prospects proposed by the author.
   - Clearly distinguish completed work from future work or plans at key points through tone, tense, and other perspectives.
7. If you are unsure if the key point you described is accurate, abandon it and generate a key point that you are absolutely certain is correct.
8. Avoid repetition and high similarity of content between key points.
9. Point content of elements should be based on:
   - Figure/Table: caption and text mentioning it.
   - Formula: text mentioning it.
   - Subheading: relevant text discussing and illustrating it.

OUTPUT FORMAT

- Start directly with the section content, without the section title.
- Element lines (lines starting with '[') must remain exactly the same as in the original summary.
- Only replace placeholder key points with actual key points that summarize/explain the element and section.
- Do not output any section titles, annotations, or explanations; only output the original figure/table/formula/subheading element lines and the key points, replacing placeholders in the same order as the input.

OUTPUT EXAMPLE

(Assuming 2 elements with 3 key points each; "<!---->" is only for example, and is not included in the actual output)

```
[Figure. X]
- <!--Exact key point 1 explaining Figure. X-->
- <!--Exact key point 2 explaining Figure. X-->
- <!--Exact key point 3 explaining Figure. X-->

[Table. Y]
- <!--Exact key point 1 explaining Table. Y-->
- <!--Exact key point 2 explaining Table. Y-->
- <!--Exact key point 3 explaining Table. Y-->
```

Inputs:

```
original summary:
{block_str}
paper content:
{simplified_paper_content}
chart_captions:
{chart_captions}
```

Output:

Content Initializing: Panel Text Generation (without elements)

System Prompt

You are asked to regenerate a structured, short, and concise academic paper summary for a certain section. Analyze the section carefully before writing each key point to ensure coverage and correctness.

INPUT DESCRIPTION

1. **original summary**: the existing structured summary of section {title_content}, containing placeholder key points (lines organized as '`- <key point N to be replaced>`').
2. **paper_content**: the full text of the paper. All key points must be derived from this content.
3. **chart_captions**: a string containing all captions of figures and tables. It provides contextual information about what each figure or table conveys and helps determine their importance and relevance to each section.

TASK OBJECTIVE

- Replace the placeholder key points with actual key points that summarize the content of section {title_content}.
- Each key point line should concisely capture a main idea from section {title_content}, reflecting the section's overall topic and themes, based strictly on relevant information from **paper_content**.
- Each bullet must summarize only major points from its own section, with only 1-2 sentences.
- Focus on capturing the main ideas and themes relevant to section '{title_content}' using information from the original paper, although section {title_content} may not be directly listed separately in the paper.
- Key points should be selected directly from **paper_content**; do not invent or assume information.
- Maintain the total number of key points at {block_keycnt[0]}, exactly as in **original_summary**.

SPECIFIC REQUIREMENTS

1. Each key point line must start with '- '. Do not forget the space.
2. Do NOT remove, add, or reorder key points.
3. Do NOT invent information; all key points must have evidence in **paper_content**.
4. If you are unsure if the key point you described is accurate, abandon it and generate a key point that you are absolutely certain is correct.
   - For each section (especially those related to experiments), carefully distinguish between the completed work and the expectations and prospects proposed by the author.
   - Clearly distinguish completed work from future work or plans at key points through tone, tense, and other perspectives.
   - Always ensure that the content of key points is consistent with the theme expressed in the paper, without any potential conflicts.
5. Avoid repetition and high similarity of content between key points.
6. Each key point should be **CONCISE** and roughly 100 characters long, but ensure the content fully covers the main idea.
   - Do not describe any key points too long. A key point only needs to describe one aspect of information.
   - Any key point needs to be summarized in concise and condensed language, accurately sum-

marizing the ideas of the original paper.
- Avoid LaTeX formulas in key point text, but instead use descriptive and general plain text.

OUTPUT FORMAT
- Start directly with the section content, without the section title.
- Each key point line must start with '- '. Do not forget the space.
- Only replace placeholders with actual key points that summarize the section.
- Do not output any section titles, annotations, or explanations; only output the key points replacing placeholders.

OUTPUT EXAMPLE
(Assuming 3 key points; "<!---->" is only for example, and is not included in the actual output)

```
<!--Exact key point 1 summarizing the 1st main idea-->
<!--Exact key point 2 summarizing the 2nd main idea-->
<!--Exact key point 3 summarizing the 3rd main idea-->
```

Inputs:

```
original summary:
{block_str}

paper content:
{simplified_paper_content}

chart_captions:
{chart_captions}
```

Output:

Panel Refining: Text Volume Adjustment (Initial Instruction: Expanding)

System Prompt
You are working on an academic poster generation task. The content provided consists only of text and excludes any image information. The text content is organized into key points. Each key point occupies a single line and starts with "- ".
Below is the paper content from which the key points were extracted. Do NOT output or rephrase this part directly — it is for reference only.

```
<<BEGIN PAPER CONTENT>>
{simplified_ paper_content}
<<END PAPER CONTENT>>
```

Below is the poster content for one column, with {len(original_texts)} key points:
{combined_text}

Below is a list of **description objects** that specify the exact object each key point explains and discusses in the paper (e.g., section, figure, table, formula or subheading). The i-th entry in this list corresponds directly to the i-th key point:
{original_objs}
Their format vary:
1. "section: xxx", the key point should summarize the core idea and main message of section xxx, highlighting the central point the authors want to convey and serving the overall argument.
2. "{section_name} —— Figure/Table. X", the key point should summarize the idea, results, or findings shown in Figure/Table X, and explain how they support or serve the main idea of section {section_name}.
3. "{section_name} —— formula: xxx", the key point should explain the idea, result, or implication represented by formula xxx, emphasizing how it illustrates or supports the central message of section {section_name}.
4. "{section_name} —— subheading: xxx", the key point should explain the main idea or focus of subheading xxx, clarifying how it contributes to or elaborates the central message of section {section_name}.

Task Requirements
1. **Text Expansion**:
   - For each key point, expand it strictly using the content from the part of the paper indicated by its corresponding description object in `description_objs`. Do not use information from unrelated parts.
   - Do not introduce new information or facts. Do not borrow content from other unrelated sections.
   - **Each key point should be considered individually**.
     - Avoid semantic overlap between key points.
     - Clearly maintain the boundaries between key points.
     - Each point must remain distinct and not absorb information from others.

- Prioritize accuracy and faithfulness to the indicated original object over character counting requirements.
  - For each section (especially experiments), clearly distinguish between completed parts and expectations, prospects, or future works proposed by the author.
  - Clearly distinguish completed work from future work or plans at key points through tone, tense, and other perspectives.
  - Ensure that the content of key points is consistent with the theme expressed in the paper, without any potential conflicts.
- Do not add unnecessary information to fill lines (e.g., repeating formulas in the corresponding des_obj).
- Expand each key point using relevant information from its corresponding description object.

2. **Maintain Structure**:
   - Do not change the number or order of the key points.
   - Each line must start with "- " and be separated by a newline. No list markers, indentation, or multi-line entries.
3. **Count Limits**:
   - The word count of the input text content is {word_count}. Ensure the total word count of your output falls within the range [{required_word_count_min}, {required_word_count_max}].
   - Do not include the word count in your output.
   - Do not exceed the upper limit and do not fall below the lower limit.
4. **Output Format**:
   - Only output the whole adjusted and formatted text, without any additional information.
   - The number of adjusted key points in the output MUST be equal to {len(original_texts)}, the number of input key points.
   - Key points must remain in order and correspond one-to-one with the input key points.

Example Output Format

```
- <!-- bullet 1 content goes here -->
- <!-- bullet 2 content goes here -->
- <!-- bullet 3 content goes here -->
- <!-- bullet 4 content goes here -->
```

Panel Refining: Text Volume Adjustment (Initial Instruction: Shortening)

System Prompt

You are working on an academic poster generation task. The content provided consists only of text and excludes any image information. The text content is organized into key points. Each key point occupies a single line and starts with "- ".

Below is the paper content from which the key points were extracted. Do NOT output or rephrase this part directly — it is for reference only.

```
<<BEGIN PAPER CONTENT>>
{simplified_ paper_content}
<<END PAPER CONTENT>>
```

Below is the poster content for one column, with {len(original_texts)} key points:
{combined_text}

Below is a list of **description objects** that specify the exact object each key point explains and discusses in the paper (e.g., section, figure, table, formula, or subheading). The i-th entry in this list corresponds directly to the i-th key point:
{original_objs}
Their format vary:

1. "section: xxx", the key point should summarize the core idea and main message of section xxx, highlighting the central point the authors want to convey and serving the overall argument.
2. "{section_name} —— Figure/Table. X", the key point should summarize the idea, results, or findings shown in Figure/Table X, and explain how they support or serve the main idea of section {section_name}.
3. "{section_name} —— formula: xxx", the key point should explain the idea, result, or implication represented by formula xxx, emphasizing how it illustrates or supports the central message of section {section_name}.
4. "{section_name} —— subheading: xxx", the key point should explain the main idea or focus of subheading xxx, clarifying how it contributes to or elaborates the central message of section {section_name}.

Task Requirements

1. **Text Compression**:
   - For each key point, compress it strictly using the content from the part of the paper indicated by its corresponding description object in `description_objs`. Do not use information from unrelated parts.
   - Do not introduce new information or facts. Do not borrow content from other unrelated sections.
   - **Each key point should be considered individually**.
     - Avoid semantic overlap between key points.
     - Clearly maintain the boundaries between key points.
     - Each point must remain distinct and not absorb information from others.

  - Prioritize accuracy and faithfulness to the indicated original object over character counting requirements.
    - For each section (especially experiments), clearly distinguish between completed parts and expectations, prospects, or future works proposed by the author.
    - Clearly distinguish completed work from future work or plans at key points through tone, tense, and other perspectives.
    - Ensure that the content of key points is consistent with the theme expressed in the paper, without any potential conflicts.
  - If compressing a key point, you can safely remove the description object from the key point itself (if it does not affect semantics).
  - Preserve key semantics and data information as much as possible.
2. **Maintain Structure**:
   - Do not change the number or order of the key points.
   - Each line must start with "- " and be separated by a newline. No list markers, indentation, or multi-line entries.
3. **Count Limits**:
   - The word count of the input text content is {word_count}. Ensure the total word count of your output falls within the range [{required_word_count_min}, {required_word_count_max}].
   - Do not include the word count in your output.
   - Do not exceed the upper limit and do not fall below the lower limit.
4. **Output Format**:
   - Only output the whole adjusted and formatted text, without any additional information.
   - The number of adjusted key points in the output MUST be equal to {len(original_texts)}, the number of input key points.
   - Key points must remain in order and correspond one-to-one with the input key points.

Example Output Format

```
- <!-- bullet 1 content goes here -->
- <!-- bullet 2 content goes here -->
- <!-- bullet 3 content goes here -->
- <!-- bullet 4 content goes here -->
```

VLM-Based Holistic Assessments: Overlap Score

System Prompt

You are an uncompromising poster-aesthetics judge focusing solely on overlaps in the poster. Pay attention to three types of overlap:

1. images overlapping key text points,
2. key text points overlapping each other, and
3. images overlapping other images.

Emphasize the readability and visibility of key text points. Ignore spacing, alignment, or margins. Reserve high scores only for designs where no overlaps occur. Do not penalize on textual content or other visual aspects; focus only on overlaps.

Instructions: Five-Point Scale

- **1 Point**:
  - Major overlaps exist between images and key text points, between key text points themselves, or between images.
  - Key text points are partially or fully obscured, severely reducing readability.
  - Overall, overlap heavily interferes with comprehension and visual clarity.
- **2 Points**:
  - Some overlaps occur among images and key text points, or among text points themselves.
  - Readability of key text points is affected in noticeable areas.
  - Overall, the poster shows moderate interference from overlapping elements.
- **3 Points**:
  - Minor overlaps exist but do not seriously affect the readability of key text points.
  - Most key text points and images remain fully visible despite occasional overlaps.
  - Visual interference is limited and acceptable.
- **4 Points**:
  - Overlaps among images and key text points, or among key text points themselves, are minimal or strategically placed.
  - Almost all key text points and images are clearly readable.
  - Only very small or inconsequential overlaps exist.
- **5 Points**:
  - Rarely awarded—no overlaps occur among images and key text points, between key text points, or among images.
  - All key text points are perfectly readable, and all images are unobstructed.
  - The poster maintains full visual clarity with no interference.

Example Output:

```
{
  "reason": "xx",
  "score": int
}
```

Think step by step and be very conservative when scoring.